ATLAS

A dense, glass-based interface system in two themes. Rounded glass over navy light, one chartreuse for every action, 10px tracked labels against 46px tabular figures.

TOKEN LAYERS	THEMES	CONTRAST PAIRS GATED	ELEVATION LEVELS	CASE SCREENS
2	2	34	4	3

00 CONTENTS

What this book holds

ATLAS is a product interface system, not a brand — there is no photography and no logotype beyond the mark. Its character lives in three structural devices: the index label, the glass panel, and the rule. Everything else in this book is the discipline that keeps those three working at 12–14px, all day, in both themes.

Two laws sit above everything. Components are built from the **semantic token layer only** — any raw hex in the component layer is a bug. And every measured claim is enforced: `tools/check_contrast.py` re-measures all 34 declared colour pairs in both themes on every run.

The book compresses; the repository is the authority. `showcase.html`, `components.html` and `foundations.html` carry every rule here as a working, theme-switchable page.

01 PHILOSOPHY

The three devices

01 THE INDEX LABEL

A 10px tracked mono label with a numbered accent prefix and a hairline running to the end of the column. Every block of content opens with one — it is the system's table of contents, printed in place.

04.2 RECONCILIATION

B APPROVER

02 GLASS

Panels are translucent white (5.5–12.5%) over the navy with an 18px blur, 30px on the app frame. A 1px gradient sheen along the top edge stands in for elevation; nothing casts a shadow downward.

There is no box-shadow in the system except the accent glow on a hovered primary button.

03 THE RULE

Ingest	48,210
Query	31,884
Exports	2,051

Inside glass, content separates with hairlines, never with nested cards. Figures sit in one divided panel, not separate tiles.

Tables are the primary surface, with an accent-tint hover instead of striping. The system is dense on purpose: 12–14px carries the interface, and nothing is inflated to a comfortable-looking 16px — a data product earns trust by showing more, legibly.

Three things were inherited from the reference system and everything visual is new: the density, the 1.75px base grid, and mono numerics for every figure, reference and identifier. The ice-blue, the cyan primary, the rounded chrome — all dropped.

Two layers, and the split is load-bearing

1 PRIMITIVES · PER THEME

Raw values, declared per theme — `--navy-900`, `--lime-400`, `--slate-300`. **Never referenced by a component.**

↓

2 SEMANTIC · THE ONLY LAYER COMPONENTS USE

`--ground` · `--surface` · `--border` · `--text-primary` · `--fill-accent` · `--tint-success` — role names that survive the theme flip.

↓

3 COMPONENTS · COMPONENTS.CSS

Buttons, fields, selection, tags, the ledger table, tabs, segmented, dialog, flags, readouts, empty states. Any hex here is a bug; the same markup flips themes untouched.

Why the split is law

A component that reaches past the semantic layer breaks when the theme flips. That is not hypothetical — it is what happened to the accent before it was split into `--fill-accent` and `--text-accent`.

Aliases are pointers

Short aliases (`--ink`, `--ac`, `--glass`) keep older markup working; every one is a pointer, never a value. New work uses the semantic names.

Machine-readable

`tokens.json` carries every token — colors, dimensions, durations, aliases as references — generated from `tokens.css` by `tools/build_tokens.py`. Edit the CSS, regenerate, never edit the JSON.

Enforced

`tools/check_contrast.py` re-measures all 34 declared role pairs in both themes on every run and exits non-zero under the bar.

The mark is the system drawn small

ATLAS

64 · HERO 34 · TOPBAR 20 · DENSE 16 · FAVICON

Construction

A raised-navy tile at the frame radius (14px), three contour hairlines, and one chartreuse peak whose crossbar is the contour it crosses — the index label, the rule and the accent, drawn once.

Two files

`mark.svg` is self-coloured and works on any ground; `mark-mono.svg` is currentColor and inline-only — it never ships as a standalone asset.

The lockup

Mark plus ATLAS in Gabarito 700 tracked 0.14em, the mark at cap height × 1.9. The mark never appears without the name except as a favicon.

Grounds

The tile carries its own navy, so the mark holds on dark, light, and any product surface — it is never recoloured to match a theme.

03 COLOUR

Dark — the theme it was designed in

GROUND #0A1A2F	GROUND-DEEP #061220	GROUND-RAISED · SOLID #0E2440	SURFACE · GLASS 5.5% WHITE 5.5%	SURFACE-HOVER · 8.5% WHITE 8.5%	BORDER · 17% WHITE 17%
TEXT-PRIMARY #EEF3F8 · 15.7:1	TEXT-SECONDARY #A9BACB · 8.8:1	TEXT-TERTIARY · LABELS #8698AC · 5.9:1	PLACEHOLDER #74889D · 4.8:1	FILL-ACCENT · TEXT-ACCENT #D4FF4F · 15.2:1	TEXT-ON-ACCENT #1B2405 · 14.0:1

A STATUS — TEXT · TINT · BORDER, IN THAT ORDER

● OK · 10.0:1 ● WARN · 10.5:1 ● DANGER · 6.3:1 ● INFO · 9.4:1

● ACCENT

Each status is three tokens — a text tone, a background tint at 13%, a border at 34% — so a tag, a flag and a plain line of status text all derive from one family.

The blooms behind the glass raise background luminance under the text, so every ratio above keeps headroom beyond its bar. All four text roles clear AA for normal text at 10px — the size the signature labels actually run at.

Light — a remap, not an inversion

GROUND #F2F5F8	GROUND-DEEP #E7ECF1	GROUND-RAISED · SOLID #FFFFFF	SURFACE · FROSTED 58% WHITE 58%	BORDER · NAVY 16% NAVY 16%	FILL-ACCENT #C9F53C
TEXT-PRIMARY #0D1A26 · 16.1:1	TEXT-SECONDARY #46596A · 6.6:1	TEXT-TERTIARY · LABELS #56697E · 5.2:1	PLACEHOLDER #596B80 · 5.0:1	TEXT-ACCENT · DEEP LIME #4A6606 · 6.0:1	TEXT-ON-ACCENT #1B2405 · 12.8:1

Hairlines invert

On dark, lines are white at 10–26%; white-on-white has no edge, so on light they become navy at 10–26%.

Glass opacity climbs an order of magnitude

5.5% white over pale grey reads as nothing; light glass sits at 52–90% — frosted rather than tinted — and a light panel needs its hairline to define the edge.

Status re-derives

Success drops to #10794B (5.0:1), warning to #8A5B04 (5.4:1), danger to #C0271F (5.4:1) — the first pass failed at 4.35:1 and below, and the gate now keeps it from regressing.

The boundary rule

The accent fill alone measures 1.16:1 against this ground, so the primary button's boundary is carried by `--border-accent` in both themes.

03 COLOUR

One chartreuse, split in two

A FILL — BACKGROUNDS, DARK INK ON TOP

Close month

● CLOSING

`--fill-accent` works as a fill in both themes — dark ink on it measures 14.0:1 dark, 12.8:1 light.

B TEXT — TYPE, ICONS, THE INDEX PREFIX

The index prefix **04.2**, a link like this one, an accent icon — all take `--text-accent`: chartreuse itself on dark (15.2:1), deep lime on light (6.0:1). Chartreuse as type on a pale ground is 1.3:1 — the token split exists because that failure shipped once.

They are not interchangeable

Anything colouring type uses `--text-accent`, never `--fill-accent`. The one exception is ink sitting *on* the fill, which uses `--text-on-accent`.

One accent, everywhere

Every action, link, active state, selected tab, focus ring and series-1 chart line is the same chartreuse family. A second accent hue is not a customisation — it is a different product (page 24).

The accent is the attention budget

One flagged item, one primary button per view. Spending chartreuse twice halves what both purchases bought.

Focus

2px chartreuse outline at 2px offset on every interactive element, following the element's own radius; light theme deepens it to lime-800 (7.2:1).

Proof — every role pair, measured, both themes

PAIR · VS GROUND	DARK	LIGHT	BAR
text-primary	15.65	16.09	4.5
text-secondary	8.80	6.62	4.5
text-tertiary — the 10px labels	5.91	5.16	4.5
text-placeholder	4.79	5.00	4.5
text-accent	15.16	6.01	4.5
text-on-accent · vs fill	14.00	12.77	4.5
text-success	10.03	4.97	4.5
text-warning	10.47	5.36	4.5
text-danger	6.30	5.40	4.5
text-info	9.36	7.01	4.5

PAIR · VS GROUND	DARK	LIGHT	BAR
viz-1	15.16	3.45	3.0
viz-2	9.36	7.01	3.0
viz-3	7.63	4.52	3.0
viz-4	8.76	5.42	3.0
viz-5	10.89	4.57	3.0
viz-6	10.47	5.36	3.0
focus-ring	15.16	7.19	3.0

Thirty-four pairs, WCAG 2.x relative luminance, re-measured by `tools/check_contrast.py` on every run — the table above is the gate's own output, not a claim. Motion honours `prefers-reduced-motion` (both durations collapse to 1ms); forced-colors mode goes solid and hands the palette to the system; touch targets floor at 44px.

The jump is the character

A SPLINE SANS MONO · THE NUMBERS

48,210

PB-2216 MARCH · RECONCILED

Everything numeric — figures, references, identifiers — is Spline Sans Mono, tabular, tracked −0.02em. The system's character is the jump between 10px tracked labels and 46px tabular figures, with very little in between.

−1,240.00 99.96% TRX-014021

B GABARITO · THE INTERFACE

Close the month with the numbers shown

Gabarito carries headings, body and controls at 12–14px. Weights 400–700; tight tracking (−0.03em) on display sizes, 0.1em on the 10px labels. Sentence case everywhere except the labels, which are always uppercase and always mono.

Labels are mono, headings are not

The two faces never trade jobs: Gabarito never sets a figure, Spline never sets a sentence.

No third face

Icons are the only other glyph system on a surface.

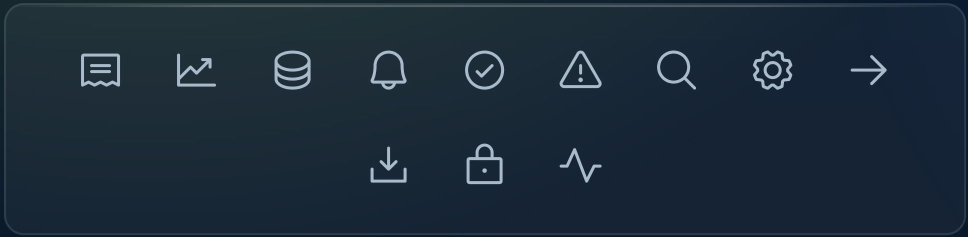
04 TYPOGRAPHY

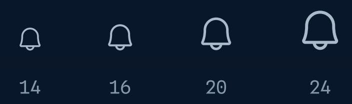
The scale, all ten steps

46	1,204,118	HERO READOUT
33	Page heading	
25	Section heading · dense KPI	
19	Panel heading	
16	Standfirst — the largest body size	
14	Large control text, comfortable body	
13	The working size — tables, controls, body. Most of the interface is this line.	
12	Dense controls, nav items, secondary cells	
11	Help text, KPI deltas, legends	
10	THE INDEX LABEL — THE SYSTEM'S SIGNATURE, AND WHY EVERY TEXT ROLE CLEARS AA AT 10PX	

10 / 11 / 12 / 13 / 14 / 16 / 19 / 25 / 33 / 46. Density never changes type — `.a-dense` tightens spacing and control heights and leaves every size above exactly where it is.

Phosphor, regular, and six rules





⚠ **The icon repeats the copy.** Severity is in the words, never only in the glyph or the colour.

1 Regular weight only

Never mixed with bold, fill or duotone — the mix is the fastest way to make an icon set look borrowed.

2 Sized in em

`.a-icon` sets 1.15em so a glyph tracks the label beside it; standalone sizes are 14 / 16 / 20 / 24px.

3 currentColor

An icon takes the colour of the text it sits with, never its own.

4 Never alone

Icon-only controls carry `aria-label`; decorative glyphs carry `aria-hidden`.

5 Never the sole carrier

A state is a tint plus a word; icons in a flag repeat what the copy already says.

6 No emoji

Anywhere. The system's warmth is chartreuse, not 🎉.

Twelve columns, four breakpoints

A THE CANONICAL SPLITS

SPAN 8 — PRIMARY	SPAN 4

3	3	3	3

Twelve columns, gutter `--sp-4` (14px — the grid's own unit, ×8). Reading surfaces cap at 1060px; application shells run to 1440px and stop — a ledger wider than 1440 stops being scannable. A 6/6 split is a smell: usually two views pretending to be one.

B BREAKPOINTS

FROM	NAME	WHAT CHANGES
0	Stack	One column. Side panels become sections below; tables shed to three columns: identity, state, figure.
768	Split	Two columns. The 8/4 split arrives; KPI rows go 2×2.
1060	Full	The twelve-column grid, KPI rows in one line, side rail visible.
1440	Cap	Nothing new appears — margins absorb the rest. Density, not width, is how ATLAS scales up.

One switch, on the region

A COMPORTABLE — DEFAULT

ENTRY	STATE	AMOUNT
TRX-014021	SETTLED	-1,240.00
TRX-014022	PENDING	-96.20
TRX-014023	SETTLED	-18,400.00

B DENSE — .A-DENSE ON THE REGION

ENTRY	STATE	AMOUNT
TRX-014021	SETTLED	-1,240.00
TRX-014022	PENDING	-96.20
TRX-014023	SETTLED	-18,400.00
TRX-014024	SETTLED	-512.75

One switch, applied to a region, never to a single control. Dense drops row padding one step and controls to 28px. **Type sizes never change** — density is spacing, not shrinking; a screen that needs smaller type needs a different screen.

Dense is for pointer surfaces; the 44px touch floor makes it unavailable on touch. Control heights run 28 / 34 / 42px, and the 34px default is itself a pointer-surface number.

06 GEOMETRY

The 1.75px unit, radii 6 / 10 / 14

A SPACE

- 3.5
- 7
- 10.5
- 14 · the gutter
- 21 · panel pad
- 28
- 42
- 56
- 84

A 1.75px base unit instead of 4 or 8. Invisible, unusual, and it works.

B RADII

6 10 14 pill

Inner elements 6, panels and controls 10, the app frame 14. Pill is reserved for tags, progress tracks and avatars — a pill-shaped button is a different design system.

C MOTION

90ms	micro — hover, press, focus, toggle. Below perception as animation; reads as response.
170ms	structural — panels, dialogs, tab content, rows entering.
one curve	`cubic-bezier(0.2, 0, 0, 1)` — fast out, long settle. Variety of easing is how motion systems rot.
1ms	reduced motion collapses both. Nothing depends on seeing the transition.

07 GLASS

Depth without a single shadow

01 A PANEL OVER LIGHT

The blur needs light behind it — `body::before` paints the blooms, and a glass panel dropped onto a flat fill reads as grey mush.

UPTIME

99.96%

P50

84ms

Acknowledge

A ELEVATION — FOUR LEVELS, LIGHT DOES THE WORK

0	ground-deep — the page behind everything
1	glass — panels, 18px blur
2	glass-lg — the app frame, 30px blur
3	solid — dialogs and menus, opaque, readable over anything

Each level catches more of the top-edge sheen; none casts anything downward. **There is no level 4** — anything above a dialog is a design error.

The caveats are part of the spec

The budget is a number

Six simultaneously visible blurred panels per view, the app frame counted once. Blur is GPU work; beyond six, group content under one panel with hairlines rather than adding panels.

The one forbidden thing

Glass never nests inside glass. Controls that live inside a glass panel — the theme toggle, the segmented control — are solid (`--ground-deep`), which is why the segmented control looks the way it does.

The fragile theme

Light glass has far less contrast against its ground than tinted white has against navy — a light panel needs its hairline to define the edge.

No backdrop-filter

`@supports` raises the fill to `--surface-active`, trading translucency for legibility.

Forced colors

Windows High Contrast gets no translucency at all: blooms off, glass solid to `Canvas`, borders to `CanvasText` — the 1px borders every panel already carries are what keeps the structure legible.

Print

Blur does not print. Paper surfaces use the solid register — this book's own pages are the demonstration: every panel above sits on an opaque bloom render, not a live blur.

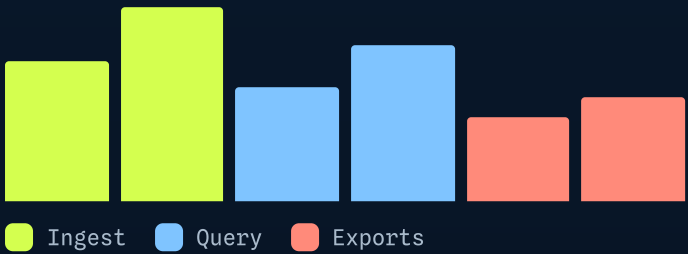
Six series, re-derived per theme

A DARK — ALL ≥ 3:1 ON THE GROUND

1 · ACCENT 2 3 4 5 6

B LIGHT — SAME SIX JOBS

1 · ACCENT 2 3 4 5 6



Order is fixed

Series 1 is always the accent, and a chart never reorders the palette to look nicer — a colour means the same series everywhere in a product.

Past six, aggregate

A seventh series is a table, not a line.

Colour is never the only carrier

A legend names every series; lines differ in weight as well as hue; trend direction pairs the colour with a sign.

The grid is quiet

`--viz-grid` is white or navy at 8–9% — reference lines never compete with data.

One signal per view, and the quiet states

RG-2214 over threshold. Flow 312 m^3/h against a ceiling of 300. Acknowledge or reassign.

One flagged item, one primary button per view. When everything is highlighted, nothing is — the danger flag above only works because it is the only red on the page. A flag inserted in response to an action carries `role="status"` or `role="alert"` so it is announced, not just painted.

● 41 NOMINAL ● 2 DRIFTING ● 1 OVER

B THE FOUR EMPTY STATES — HAIRLINE SLATE, ONE CHARTREUSE ACCENT

Actions, fields, selection

A BUTTONS · DEFAULT / PRIMARY / DANGER / GHOST · SM / MD / LG

Export | Close month | Void entry | Cancel

Small · 28 | Default · 34 | Large · 42

B FIELDS · LABEL, HELP, INVALID

A APPROVER

E. Papadaki — Controller

B THRESHOLD

300.00

Above the licensed ceiling of 250.00 — lower it or attach a variance.

C SELECTION

Reviewed | Monthly | Quarterly

A button is a button

Native `<button>`, never a styled div — Space and Enter come free and stay free. Icon-only buttons carry `aria-label`.

Invalid is an attribute

The stylesheet keys the danger border and tint to `aria-invalid="true"` — never to a class — and the error line is wired in with `aria-describedby`. The error says how to fix it.

State lives in :checked

Selection controls are native inputs inside their labels; the label text is the accessible name; radios arrow-key within their `name` group; the switch carries `role="switch"`.

The numeric field

`.a-input--num` is mono, tabular and right-aligned — a figure is a figure even while it is being typed.

The ledger, tabs, dialog, readouts

A THE LEDGER TABLE — THE PRIMARY SURFACE

ENTRY	COUNTERPARTY	STATE	AMOUNT
TRX-014021	Meridian Freight	SETTLED	-1,240.00
TRX-014022	Harbour Utilities	PENDING	-96.20
TRX-014023	Aegis Payroll	SETTLED	-18,400.00

B TABS & SEGMENTED

Statement Exceptions Audit

DARK LIGHT

Void entry TRX-014022?

The entry is reversed, the counterparty notified, and the audit trail keeps both sides. This cannot be undone.

Cancel Void entry

NET POSITION

-41,298EUR

-4.2% vs Feb

71% CLOSE PROGRESS

The contract, written down

COMPONENT	ROLES & STATES	KEYS
Button	native `<button>`; `aria-label` when icon-only	Space · Enter
Field	`label[for]` · `aria-describedby` · `aria-invalid`	—
Radio set	native inputs, shared name	↑↓←→
Switch	checkbox + `role="switch"`	Space
Tabs	`tablist / tab / tabpanel`; roving tabindex; `aria-selected`, `aria-controls`	←→ · Home · End
Segmented	`role="group"` + `aria-pressed`; one pressed at all times	Space · Enter
Dialog	`role="dialog"` · `aria-modal` · `aria-labelledby`; focus trapped, returned to opener	Esc · Tab cycles
Flag	`role="status"` (ok/info) · `role="alert"` (danger) when inserted	—
Table	`th[scope]`; sort buttons inside `th` with `aria-sort`	Space · Enter
Nav	`<nav>` + `aria-current="page"`	—
Progress	`role="progressbar"` + value min/max/now	—

ATLAS ships as CSS, so the markup and the keyboard are the implementer's to build — but not the implementer's to design. `docs/BEHAVIOR.md` is the contract, one section per interactive component, and the stylesheet keys its styling to these attributes (`aria-selected`, `aria-pressed`, `aria-invalid`, `aria-current`) — markup that honors the contract renders correctly for free, and markup that skips it renders visibly wrong, which is the point.

Focus is never removed

2px accent outline at 2px offset, on every interactive element, following the element's own radius.

Tab leaves the tablist

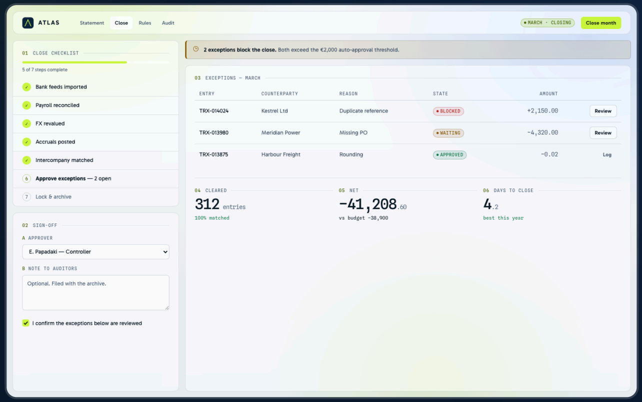
It never walks the tabs; arrow keys do. A panel with no focusable content carries `tabindex="0"`.

The overlay click may close a dialog; it is never the only way out

The footer always carries a real button.

Three products, one language

OPS CONSOLE · A SENSOR FLEET · DARK



MONTH CLOSE · A FINANCE PRODUCT · LIGHT

STATUS PAGE · A PUBLIC SURFACE · DARK

Deliberately different products, every one composed from `components.css` alone — no per-case stylesheet exists. The console leans on the one-signal rule and the data series; the close leans on forms, steps and the ledger; the status page leans on flags, uptime readouts and the solid subscribe row.

Each case is also a proof of a specific claim: the console that the density scales, the close that the same markup flips themes untouched, the status page that the language holds on a public, low-chrome surface. A new case should stress a rule, not decorate one.

10 SKINNING

Making it another product's

STEP	WHAT CHANGES
1	The accent trio — `--lime-400 / 300 / 500` — becomes the new product's signal colour.
2	The ground pairs — `--navy-900/950` dark, `--pale-100/200` light.
3	`--text-accent` is re-derived for the new accent against both grounds — the fill almost never works as type.
4	`python3 tools/check_contrast.py` — all 34 pairs must clear before anything ships.
5	Nothing else. The semantic layer and every component follow untouched.

The case screens were built exactly this way. If a reskin wants a second accent, new radii or a friendlier 16px body, it does not want ATLAS — it wants a different system, and that is a fine thing to want.

B THE PIPELINE

```
$ vi tokens.css
$ python3 tools/build_tokens.py
  tokens.json regenerated
$ python3 tools/check_contrast.py
  All 34 pairs clear. The gate is open.
```

Edit the CSS, regenerate the JSON, run the gate. `tokens.json` feeds design tools and code-gen; the gate keeps the measured claims on pages 6–9 true forever. Self-host Gabarito, Spline Sans Mono and Phosphor before production — the CDN links are a development convenience, not a dependency.

This book is generated from the living system — tokens, components and cases in the repository are the source of truth, and the contrast gate re-measures every colour claim on each run. All product data shown is invented placeholder.

Repository: `github.com/aposfys/atlas-design-system`
Source: `docs/book.html` — regenerate the PDF from it after any token change.
Live pages: `showcase.html` · `components.html` · `foundations.html` — every rule here, working, with the theme toggle.